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Complex interactions with nutrients and sediment alter the effects of predation on a reef-building coral

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Abstract

Framework-building corals create the three-dimensional structure of coral reefs and are subject to predation from fishes, echinoderms, and gastropods. Anthropogenic stressors can magnify the effects of such top-down pressure on foundation species. The gastropod *Coralliophila violacea* (Kiener, 1836) depletes coral energy reserves via predation, potentially increasing coral susceptibility to land-based pollution (i.e., sediment accumulation and nutrient pollution). We hypothesized that sedimentation would worsen coral mortality, while nutrient enrichment would mitigate the harmful effects of sediment and predation on coral mortality by increasing the densities of algal symbionts. To test these hypotheses, we conducted in situ surveys of the fringing reefs in Mo'orea, French Polynesia to explore the relationships among massive *Porites spp*. cover, *C. violacea* densities, and sediment accumulation on coral colonies across low and high nutrient sites. We also conducted a factorial field experiment to test the interactions among these stressors on coral tissue mortality, symbiont densities, and chlorophyll. Massive *Porites* colonies at higher nutrient sites had *C. violacea* densities 13 times higher than at low nutrient sites but there was no difference in the amount of live tissue on coral colonies with or without snails among these sites. In our experiment, there were interactions between predation and nutrients as well as nutrients and sediment that impacted coral mortality. Sedimentation and predation by *C. violacea* increased coral tissue mortality independently by ~20%. Nutrient enrichment reduced this effect in corals under sedimentation or predation pressure by lowering coral tissue mortality by 18% and increasing algal symbiont densities by ~28%. Our results indicate that sediment does not magnify top-down pressure on this coral, and that moderate nutrient enrichment may interact with predation in complex, unexpected ways to alter the responses of corals to top-down pressure.

KEYWORDS

corallivory, foundation species, land-based pollution, nutrient pollution, scleractinian, sedimentation

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1 | INTRODUCTION

Foundation species create important habitat that supports high biodiversity across many ecosystems (Dayton, 1972; Ellison, 2019). However, habitat destruction, global climate change, and land-based pollution have driven unprecedented declines of foundation species over the past several decades, particularly in coastal ecosystems (Fabricius, 2005; Gedan et al., 2009; Hughes et al., 2018; Lotze et al., 2006). Anthropogenic stressors often occur simultaneously and may modify one another or biotic interactions, such as predation or herbivory, to exacerbate the loss of these foundation species in coastal regions (Silliman et al., 2013). For example, in salt marsh ecosystems, drought induced by climate change can act synergistically with herbivory by the snail *Littorina irrorata* (Say, 1822) to accelerate the decline of key salt marsh plant populations, resulting in widespread loss of a foundation species (Silliman et al., 2005). The reduction in population sizes of foundation species in coastal ecosystems can result in a cascade of biodiversity loss (Angelini et al., 2011) and the subsequent alteration of ecosystem services such as nutrient cycling and the buffering of erosion (Worm et al., 2006). Understanding the potential interactions among anthropogenic stressors and biotic interactions on foundation species is critical to determining the trajectory of community structure for coastal ecosystems in the Anthropocene.

In the past several decades, the abundance of scleractinian corals on tropical reefs has declined significantly in many regions due to a host of anthropogenic stressors ranging from global climate change (e.g., rising sea surface temperatures and ocean acidification) to local stressors such as overfishing and land-based pollution (Hoegh-Guldberg et al., 2017; Hughes et al., 2017). Land-based pollution from coastal development, sewage, agriculture, and dredging can expose corals to sediment accumulation and nutrient enrichment (Fabricius, 2005; Lapointe et al., 2019), often resulting in the local loss of corals (Fabricius, 2005; Rogers, 1990; Wear & Thurber, 2015). Sediment in the water increases turbidity, which limits photosynthesis and growth rates, suppresses coral recruitment, and increases the mortality of adult corals (Burke et al., 2002; Fabricius, 2005; Rogers, 1990). Sediment accumulation on coral surfaces reduces energetic resources of corals as they expend energy to shed sediment particles (Sheridan et al., 2014). Prolonged sediment accumulation can also kill coral tissue (Weber et al., 2006), increasing overall coral mortality (Erftemeijer et al., 2012). Nutrient enrichment, another consequence of land-based pollution, can decrease coral calcification (Shantz & Burkepile, 2014) and promote coral bleaching and disease (Vega Thurber et al., 2014). Although sedimentation and nutrient enrichment often have negative effects on corals in isolation, the few studies that have examined the combined effects of these stressors on corals surprisingly have found no interactive effects on coral cover, survival, or growth (Gil et al., 2016; Muthukrishnan & Fong, 2014). Consequently, the effects of sedimentation and nutrient enrichment on corals may be more prominent in isolation than when these stressors co-occur.

Even without anthropogenic stressors, natural biotic interactions can impair coral physiology, hinder coral growth, and increase tissue mortality. Corals are subject to predation (corallivory) by a wide range of taxa, including fishes, echinoderms, molluscs, and arthropods (Cole et al., 2008; Rice et al., 2019; Rotjan & Lewis, 2008). Browsing corallivory by gastropods can have substantial impacts on coral health by removing coral tissue without impacting the coral skeleton. In the Caribbean and Pacific Oceans, *Coralliophila* spp. snails can significantly increase coral mortality (Knowlton et al., 1990; Raymundo et al., 2016; Shaver et al., 2017; Williams & Miller, 2012). These predatory snails feed by permeating coral tissues with a proboscis to consume photosynthate products produced by algal symbionts (Ward, 1965). This feeding mode creates an energy "sink" as the coral colony translocates carbon products to the site of injury, depleting energy reserves elsewhere in the coral (Oren et al., 1998). The feeding strategy employed by *Coralliophila* spp. reduces coral growth substantially (Clements & Hay, 2018; Hamman, 2018, 2019), alters coral morphology (Hamman, 2019), and accelerates tissue loss and mortality (Raymundo et al., 2016; Shaver et al., 2017). Thus, corallivores have substantial effects on corals at the individual and population level.

Corallivory may also compromise the ability of corals to cope with concurrent abiotic stressors (Rice, Ezzat, et al., 2019), such as sedimentation and nutrient pollution. For example, sediment exposure can deplete coral energy reserves (Sheridan et al., 2014) and inhibit tissue recovery following predation (Cróquer et al., 2002). In fact, simulated parrotfish and butterflyfish wounds on colonies of *Orbicella annularis* (Ellis and Solander, 1786) healed 2–3 times slower at sites with high sediment loads than at sites with low sediment loads (Cróquer et al., 2002). Anthropogenic sources of nutrient enrichment can also reduce the wound healing capacity of corals (Koop et al., 2001; Renegar et al., 2008). However, nutrient enrichment does not always mediate how predators influence coral growth or mortality (Shaver et al., 2017). Indeed, moderate levels of nutrient enrichment can have positive effects on corals (e.g., increased zooxanthellae densities and coral growth) (Ezzat et al., 2015; Gil, 2013). Given that sedimentation and nutrient enrichment can positively covary on nearshore reefs (Fabricius, 2005), these concurrent stressors may interact in complex ways to alter the impact of corallivory on corals.

Here, we test the potential interactive effects of sedimentation, nutrient enrichment, and corallivory on the physiology and mortality of a common reef-building coral in the tropical South Pacific. Specifically, we used field surveys to test the following hypotheses about the spatial relationships among corallivory, nutrients, and sediment: (i) Nutrient availability and sediment accumulation on massive *Porites* colonies will be positively correlated across reef sites, (ii) there will be a lower massive *Porites* cover at high nutrient sites, and (iii) there will be more corallivorous snails at sites with higher nutrient availability. We then used a factorial field experiment to examine the following hypotheses about the impacts of corallivory, nutrients, and sediments on corals: (i) The effects of nutrient enrichment and sedimentation will be antagonistic on coral mortality and physiology (i.e., enhanced algal symbiont physiology for enriched corals counterbalances impaired symbiont physiology under sedimentation), (ii) sedimentation will exacerbate the negative effects of snails on coral

mortality and algal symbiont physiology, and (iii) nutrient enrichment will lessen the impact of snail predation on coral mortality by increasing algal symbiont densities for enriched corals.

2 | METHODS

2.1 | Study site and species

This study was conducted in Mo'orea, French Polynesia (17°28.770′S, 149°47.618′W), site of the Richard B. Gump South Pacific Research Station. Mo'orea is a high volcanic island surrounded by a barrier reef composed of fringing- and back-reef habitats within the lagoon and a fore reef seaward of the reef crest. The island experiences a cooler dry season (May – October) and a warmer wet season (November – April). Heavy rainfall during the wet season can inundate portions of the fringing reefs with anthropogenic, land-based runoff, resulting in pulses of sediment that cover corals and increase nutrient concentrations in nearshore waters (Adjeroud & Salvat, 1996; Schrimm et al., 2002).

We focused our study on the common reef-building coral massive *Porites* because it is a dominant coral in the Mo'orea fringing reefs yet has been in decline in these habitats over the past decade (Edmunds, 2020). In Mo'orea, the lagoon is dominated by several species of massive *Porites* with *Porites lutea* (Edwards, 1,860) and *Porites lobata* (Dana, 1846) being the most common (Bosserelle et al., 2014). Distinguishing among these species by corallite morphology is challenging in situ, and corallite morphology of *Porites* spp. does not always conform to distinct genetic clades (Forsman et al., 2009). Thus, we focused on massive *Porites* colonies that shared common color and morphology. Massive *Porites* are frequently preyed upon by *Coralliophila violacea* (Kiener, 1836) (McClanahan, 1994), which is chemically attracted to massive *Porites* (Hamman, 2018). This corallivorous gastropod feeds on a single coral colony for extended periods of time, impairing the physiological performance of the colony and increasing tissue loss and mortality (Clements & Hay, 2018; Hamman, 2018, 2019; Oren et al., 1998; Raymundo et al., 2016).

2.2 | Field surveys of snails, nutrient availability, and sedimentation across reefs

To assess the spatial relationships among corallivorous snails, nutrient enrichment, and sedimentation at the reef scale, we conducted in situ surveys across seven fringing reefs at ~1–2 m depth on the north shore of Mo'orea (Figure 1a). In preliminary surveys, *Coralliophila violacea* was only observed on massive *Porites* colonies >10 cm in diameter. Thus, we surveyed all massive *Porites* colonies >10 cm in diameter found in 25 × 2 m belt transects parallel to shore (n = 4 transects per site). For each colony, we recorded the flat colony diameter parallel to the benthos to the nearest cm, estimated percent live tissue of the coral colony, and counted the number of *C. violacea* snails found on each colony. Colony diameter was used to estimate the colony surface area using the surface area of a hemisphere ($2\pi r^2$), which we found closely resembled the shape of all the massive *Porites* colonies in our study. This area was then multiplied by the estimated value for percent live tissue to estimate the surface area of live tissue.

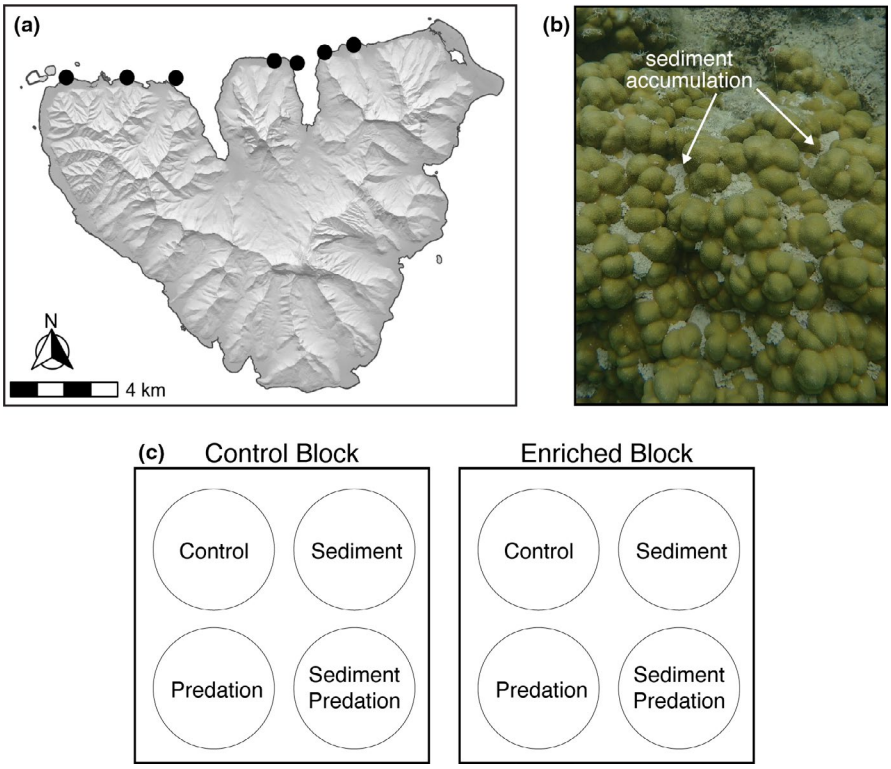

FIGURE 1 (a) Site map of the in situ fringing reef survey sites (black points) in Mo'orea, French Polynesia. (b) Example of sediment accumulation on massive *Porites* colonies in Mo'orea. (c) Schematic of the design for the field experiment testing interactions among snail predation, nutrients, and sedimentation (n = 11 massive *Porites* fragments per treatment). Note that the nutrient treatment was established at the cinderblock level and the snail predation and sediment treatments were applied within each cinderblock

Benthic photograph quadrats (25 × 25 cm) were taken on alternating sides at one-meter intervals on each transect using an Olympus TG-4 camera to quantify massive *Porites* cover at the sites. These photograph quadrats were processed in the CoralNet Software using the stratified random point distribution setting with 50 points per frame and a confidence threshold of 95% for computer automation (n = 1,402 photographs total) (Beijbom et al., 2015). These same photograph quadrats were used to determine sediment prevalence on massive *Porites* colonies using the "Fiji" version of ImageJ software (Schindelin et al., 2012). To keep the colony size class consistent with those used from our in situ surveys, each massive *Porites* colony >10 cm diameter was evaluated for presence or absence of sediment on the colony (i.e., sediment accumulation) (Figure 1b). For each occurrence of sediment accumulation on massive *Porites*, we noted if the coral tissue adjacent to the sediment was pale as this is often an indicator of physiological stress.

We used *Turbinaria ornata* (Turner J. Agardh, 1848), a brown macroalga, as a proxy for seawater nutrient concentration at these sites because algae can integrate nutrient availability in seawater over long time periods (~3 months) (Atkinson & Smith, 1983; Lin & Fong, 2008). Both observational and experimental studies show the nutrient levels within algal tissues are elevated when algae occur in nutrient-enriched environments relative to nutrient-poor environments (Burkepile & Hay, 2009; Lin & Fong, 2008; Vega Thurber et al., 2014). In Mo'orea, nitrogen content of *T. ornata* is correlated with anthropogenic sources of nutrients (Adam et al., 2020). Samples of *T. ornata* (n = 10 fronds per site) were collected at each site, cleaned of epiphytes, and dried at 60°C for ~48 hr. These samples were then ground to a fine powder and processed for the percent carbon, hydrogen, and nitrogen (CHN) via elemental analysis using an Automated Organic Elemental Analyzer (Exeter Analytical CEC440HA) in the analytical laboratory of the University of California Santa Barbara Marine Science Institute. *Turbinaria ornata* in enriched environments often have tissues with higher than 0.5% N (Schaffelke, 1999), enabling us to categorize each survey site as either a low (N < 0.5%) or high nitrogen site (N > 0.5%). The % N of *T. ornata* tissues was significantly higher at the high nitrogen sites (N > 0.5%) than the low nitrogen sites (N < 0.5%) (Mann–Whitney; p =.02; range: 0.45%–0.62% N; Figure S1).

2.2.1 | Analysis of field survey data

Differences in *Coralliophila violacea* densities at low and high N sites were analyzed with a Mann–Whitney test because the data did not follow a Gaussian distribution. Chi-squared analysis was used to evaluate differences in the prevalence of sediment accumulation on massive *Porites* between low and high N sites. Differences in the percent live tissue of massive *Porites* colonies with and without snails were evaluated with Mann–Whitney tests within low and high N sites. A linear regression was used to evaluate the relationship between snail density and percent tissue mortality for massive *Porites* colonies with snails. All analyses were conducted in R (v3.4.3) (R Development Core Team, 2017), and figures were produced using the *ggplot2* and *ggthemes* packages (Arnold, 2017; Wickham & Wickham, 2009).

2.3 | Field experiment testing interactions of snails, nutrient availability, and sedimentation on massive *Porites*

To evaluate how predation by *Coralliophila violacea* interacts with nutrient enrichment and sedimentation to affect massive *Porites*, we conducted a six-week (July–September 2017) full-factorial study at a back-reef site on the north shore of Mo'orea. We conducted the field experiment at a back-reef site because these sites are characterized by less sedimentation and nutrient loading relative to the fringing reef sites in Mo'orea, which allowed us to disentangle the effects of these stressors on corals. Eight fragments (17.6 ± 0.7 cm^2 surface area) were collected from 11 massive *Porites* parent colonies at the back-reef site using a hammer and chisel (88 fragments total). We chose apparently healthy parent colonies without any signs of predation or tissue loss. Coral fragments were immediately returned to the Gump Research Station by boat after collection. Each coral fragment was attached by epoxy adhesive (Z-spar A-788) onto the open neck of a 50 ml Falcon tube and allowed to acclimate for ~24 hr in flow-through seawater tables. Inverted 50-ml Falcon tube caps were epoxied to 20 × 20 cm cinderblocks (n = 4 caps per cinderblock) deployed at the back-reef site. This method allowed us to easily deploy coral fragments in the field (see Clements & Hay, 2015 for methods). Four massive *Porites* fragments were attached to each cinderblock. Each cinderblock was covered with an enclosure of wire mesh (2.5 × 2.5 cm diameter holes) to prevent fish predation on corals. Cinderblocks were placed ~1.5–2 m apart in a single row running parallel to shore.

Corals were randomly assigned to the following treatments (n = 11 per treatment; see Figure 1c): (i) control, (ii) snail predation, (iii) nutrient enrichment, (iv) sediment addition, (v) snail predation and nutrient enrichment, (vi) snail predation and sediment addition, (vii) nutrient enrichment and sediment addition, and (viii) snail predation, nutrient enrichment, and sediment addition. Neither initial coral mass, determined by the buoyant weight method (Davies, 1989; Jokiel et al., 1978), nor coral surface area, measured using the aluminum foil method (Marsh, 1970), varied across treatments (one-way ANOVA; mass: $F_{7,84}$ = 0.51, p =.83; 12.3 ± 0.7 g, mean ± *SE*; surface area: $F_{7,79}$ = 1.25, p =.28; 17.7 ± 0.7 cm^2). The fragments of massive *Porites* acclimated to field conditions at ~2 m depth for ~48 hr before treatments began.

Nutrient treatments were established at the cinderblock scale with every other cinderblock in the row assigned to the enriched treatment. For enriched blocks, we affixed two nutrient diffusers to opposite sides of each cinderblock. Nutrient diffusers were 10 cm long ×2.5 cm wide PVC pipe with 10 0.6 cm holes drilled into the sides. These pipes were enclosed in window screen mesh and

then filled with 30 g slow-release Classic Osmocote (19–6–12 of N-P-K). This method has been successful in elevating the nutrient concentrations of seawater in previous studies (Bruno et al., 2003; Burkepile & Hay, 2009; Gil et al., 2016; Shaver et al., 2017). Nutrient diffusers were changed every 2–3 weeks to maintain elevated nutrient levels throughout the duration of the experiment. We collected seawater samples ~5 cm above the surface of each cinderblock before each diffuser swap using acid-washed 20-mL syringes. These seawater samples were kept on ice while transported by boat back to the Gump Research Station within <1 hr of collection. All samples were immediately filtered through 0.7 μm GF/F filters into acid-washed vials and frozen at −40°C until analysis. A subset of these samples was processed for $NO_2^- + NO_3^-$ and PO_4^{3-} by the Cochlan Laboratory at San Francisco State University. Frozen samples were thawed overnight at room temperature (~20°C) prior to analysis. For the analysis, a Lachat Instruments flow injection automated ion analyzer (8000 Series; Hach Co.) was used following the methods of Smith and Bogren (2001) for NO_3+ NO_2 and Knepel and Bogren (2002) for PO_4.

Sediment treatments were established on two coral replicates within each cinderblock to mimic the sediment regime of Mo'orean fringing reefs where sediments are <500 μm (~99% of collected sediments) (Gil et al., 2016). To simulate this sediment regime, we filtered 5 ml sediment from the sandy sea floor adjacent to each cinderblock through a 500-μm nylon-mesh sieve to cover the surface of each sediment treatment coral replicate with sediment particles twice per week. This resulted in a maximum sedimentation rate of ~100 mg cm^2 day^{-1} as a large portion of the sediment particles fell onto the cinderblock or were not filtered through the sieve due to particle size. This sedimentation rate falls toward the high end of the range of sediment deposition observed on coral reefs (see table S1 in Gil et al., 2016). To evaluate whether the sedimentation treatment increased sediment cover on coral fragments, photographs were taken before and immediately after sediment addition for each coral fragment and a uniform 100-point grid was superimposed on each photograph to determine the percent change in sediment cover before and after sediment addition. Corals were exposed to nutrient enrichment and sediment addition treatments for ~2 weeks before the predation treatment began.

Snail predation treatments were also established within each cinderblock. We collected *Coralliophila violacea* (13.9–26.0 mm shell length, 16.87 ± 0.32 mm mean shell length) from the same back-reef site as the experiment and brought the snails back to the Gump Research Station to measure shell length to the nearest 0.01 mm using digital calipers. Following Clements and Hay (2018), snails were attached to massive *Porites* fragments using cable ties. The shell of each snail was sanded with sandpaper, glued to a cable tie with cyanoacrylate adhesive, and attached to respective corals with one snail per coral for the predation treatments (Clements & Hay, 2018). Each snail was allowed to feed on an individual massive *Porites* fragment for 10 days before being removed, which is sufficient time for predation wounds by *C. violacea* to be visible on *Porites* corals (Raymundo et al., 2016). Sediment was not added to any coral fragments during the predation period because it can deter snail feeding (Suren, 2005). After the 10-day predation period, the wounds were monitored for signs of healing (e.g., polyp regeneration) for ~2 weeks as *Porites spp*. heals most snail predation wounds within this time frame (Raymundo et al., 2016). Photographs of the predation scars, including a ruler for scale, were taken with an Olympus TG-4 camera immediately following snail removal and again at the end of the experiment. Wound healing rates were determined by tracing the snail predation scar in ImageJ in the initial and final photographs. The percent tissue mortality for each coral was estimated visually during weekly in situ surveys on SCUBA by the same observer (M. Rice) throughout the experiment.

Water temperature (HOBO Pendant loggers; ±0.53°C accuracy) and photosynthetically active radiation (PAR; Odyssey) were recorded at ~2 m depth throughout the experiment. PAR sensors were calibrated according to manufacturer specifications, mounted with cable ties to a cinderblock, and cleaned weekly with brushes to prevent biofilm accumulation. Sedimentation rates at the site were recorded at several locations using simple sediment traps made of PVC tubes 5 cm in diameter with an aspect ratio of 5:1 (White, 1990). Sediment was collected weekly, dried at 60°C for ~48 hr, weighed to the nearest 0.01 g, and converted to sedimentation rates (mg cm^{-2} day^{-1}) using the inner diameter of the PVC tube.

2.3.1 | Densities of Symbiodiniaceae and chlorophyll production in massive *Porites*

At the end of the experiment, we analyzed coral tissue from all live massive *Porites* fragments to evaluate treatment effects on algal symbiont densities and chlorophyll *a* production, as these metrics are proxies for the physiological health of corals. Coral tissue was removed from each fragment using an airbrush and 0.7 μm filtered seawater (FSW) into a 15 ml Falcon tube. The total volume of coral tissue slurry was recorded to the nearest 0.1 ml for each fragment. The coral tissue slurry was homogenized using a homogenizer (Fisherbrand 150 Hand Held Homogenizer Motor) with a stainless-steel saw tooth probe for 10 s, and 1 ml of the homogenous coral tissue slurry was aliquoted to a 2 ml microcentrifuge tube for chlorophyll *a* analysis. The remaining tissue slurry was spun down at 4,000 rpm for 5 min, and the supernatant was discarded. This step was repeated three times to ensure the complete removal of residual tissue. The Symbiodiniaceae (formerly *Symbiodinium*; LaJeunesse et al., 2018) pellet was diluted with 10 ml of FSW. Symbiodiniaceae densities were analyzed using a hemocytometer and compound microscope (n = 4 counts per coral replicate) and normalized by coral surface area using the aluminum foil method (Marsh, 1970). Chlorophyll *a* content was determined by spinning down the homogenized tissue aliquot, removing the supernatant, and adding 5 ml 100% acetone (Jeffrey & Humphrey, 1975). Samples were kept in the dark at 4°C for 24 hr and then measured at 663 and 630 nm using a spectrophotometer. Chlorophyll *a* was calculated using the equation

for chlorophyll *a* in dinoflagellates from Jeffrey and Humphrey (1975) after accounting for homogenate and solvent volumes and normalized by symbiont cell count.

2.3.2 | Analysis of factorial field experiment

A one-sided Mann–Whitney test was used to assess differences in the nutrient concentrations of our control and enriched cinderblocks because data did not follow a Gaussian distribution. To evaluate the effectiveness of our sediment addition treatment, we used a one-way ANOVA to determine the difference in the percent cover of sediment across treatments. We evaluated the impact of the snail predation treatment using a one-way ANOVA to determine whether the size of predation wounds varied across the four predation treatments: (i) predation, (ii) nutrient enrichment and predation, (iii) sediment addition and predation, and (iv) nutrient enrichment, sediment addition, and predation.

Linear mixed-effects models (LMMs) were used to evaluate the treatment effects (predation, nutrients, sedimentation, and their interactions) on tissue mortality, Symbiodiniaceae densities, chlorophyll *a* content, and the wound healing rate of massive *Porites* corals in the factorial field experiment. The *lmer* function from the *lmerTest* package in R was used to create all LMMs (Kuznetsova et al., 2017). Initially, all LMMs included parent colony and cinderblock as random effects fit with random intercepts to account for variation within the experimental design. Akaike's information criterion for small sample size (AICc) was used to test for the goodness of fit for the random effects structure using the *AICc* function from the *MuMIn* package (Barton & Barton, 2019). Models with ΔAICc of <2 are considered equivalent model fits (Burnham & Anderson, 2002), and in this case, the simplest model with the lowest AICc value was considered the best fitting model. Here, we only report the random effects that were included in the final models. The treatment effects (nutrient enrichment, sedimentation, and snail predation) were included as crossed fixed effects in all LMMs to test a priori hypotheses. Snail shell length was included as a fixed effect in all LMMs because we expected larger *Coralliophila violacea* snails to have a more negative influence on corals and wanted to account for this variation in the models. The final model structure of the fixed and random effects for each response variable is presented in Table S1.

Treatment effects on the tissue mortality of massive *Porites* were assessed with a LMM after being log (x + 1) transformed with predation, nutrient, and sediment and their interactions as fixed effects with snail shell length as a fixed effect. Cinderblock and survey date (repeated measures component) were included as random effects. Prior to snail addition, there was little tissue mortality (~0.3% average across treatments). Only data from after the snail addition were included in this analysis so that we could assess the interactions of all three stressors simultaneously. We used independent LMMs to evaluate the treatment effects on Symbiodiniaceae densities and chlorophyll *a* concentrations with predation, nutrient, sediment, and their interactions as fixed effects as well as snail shell length as a fixed effect and parent colony as a random effect. Chlorophyll *a* was log (x + 1) transformed to meet the assumptions of normality and homoscedasticity. The percent of the predation wound that healed was analyzed with a LMM with nutrient, sediment, and their interactions as fixed effects as well as snail shell length as a fixed effect with cinderblock and parent colony as random effects. All code and raw data used for this study are available at https://github.com/mallorymrice/SNAIL_2017.

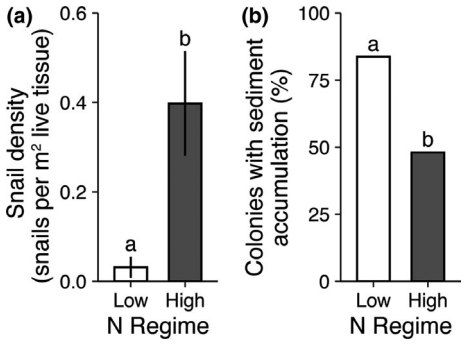

FIGURE 2 (a) The density of *Coralliophila violacea* snails on massive *Porites* colonies (snails per m^2 live tissue; mean ± *SE*; n = 426 colonies) observed across sites with low and high nitrogen (N) regimes (white bars: low N site; gray bars: high N sites). (B) The percent of massive *Porites* colonies with sediment accumulation across site N regimes (n = 120 colonies). The nitrogen levels in tissue samples of the macroalgae *Turbinaria ornata* served as a proxy for nutrient enrichment at the sites and were used to distinguish survey sites as either low (N < 0.5%) or high N sites (N > 0.5%). Letters on the bars indicate significant differences according to a Mann–Whitney test and chi-square test, respectively

3 | RESULTS

3.1 | Field surveys of snails, nutrient availability, and sedimentation across reefs

We observed 426 massive *Porites* colonies (>10 cm diameter) in our in situ surveys, and ~5% of these colonies had *Coralliophila violacea* snails (n = 23 colonies with snails; range: 1–20 snails per colony). The density of *C. violacea* was an order of magnitude higher at sites characterized by high N (0.4 ± 0.1 snails per m^2 live coral tissue) than low N (0.03 ± 0.02 snails per m^2 live coral tissue) (Mann–Whitney; p < .001; Figure 2a). For the colonies with snails, the mean density of snails observed was 4.69 ± 0.95 snails per m^2 live coral tissue. The percent of live tissue for massive *Porites* colonies did not differ for colonies with and without *C. coralliophila* at low or high N sites (Mann–Whitney; p = .5 and p = .4, respectively; Figure 3). Colonies with snails displayed no relationship between snail density and percent live tissue (linear regression; $F_{1,21}$ = 1.53, p = .23, R^2 = .02).

There were 120 massive *Porites* colonies (>10 cm diameter) in the photograph quadrats used to assess benthic cover, and ~60% of these colonies had sediment accumulation on their live tissue

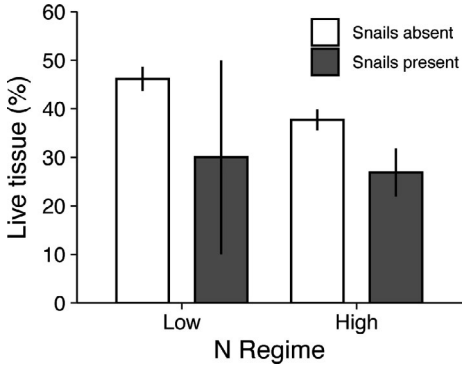

FIGURE 3 The percent live tissue of massive *Porites* colonies (mean ± *SE*) with snails absent (white bars, $n = 403$) and snails present (gray bars, $n = 23$) across low and high site nitrogen (N) regimes. Note the high variability for colonies observed with snails at low N sites is due to low sample size ($n = 2$). Tissue samples of the macroalgae *Turbinaria ornata* were used as a proxy for nutrient supply to characterize survey sites as either low (N < 0.5%) or high N sites (N > 0.5%)

TABLE 1 Linear mixed-effects model results for percent partial colony mortality during the course of the experiment with Kenward–Roger approximation for degrees of freedom

Fixed effects	*df*	*F*	*p*
Predation	1	2.732	.0997
Nutrient	1	1.033	.322
Sediment	1	0.178	.674
Snail shell length	1	1.385	.241
Predation × Nutrient	1	7.337	**.00730**
Predation × Sediment	1	4.530	**.0344**
Nutrient × Sediment	1	3.796	.0527
Predation × Nutrient × Sediment	1	0.341	.560

Notes: *p*-Values defined as significant at a threshold of .05 are highlighted in bold.

surfaces with 80% of these occurrences associated with paling tissue. The percent of massive *Porites* colonies with sediment accumulation was higher at low N sites than high N sites ($\chi^2 = 13.27$; $p < .001$; Figure 2b). Massive *Porites* cover was ~2% of total benthic cover across sites and represented ~25% of total coral cover.

3.2 | Field experiment testing interactions of snails, nutrient availability, and sedimentation on massive *Porites*

Throughout the experiment, seawater temperature at the site ranged between 24.6 and 30.5°C and had an overall mean of 27.3 ± 0.5°C (±*SE*). PAR had an overall mean of 1,285.1 ± 17.0 μmol m^{-2} s^{-1}. Sedimentation rates at the study site averaged 0.61 ± 0.087 mg cm^{-2} day^{-1}. We were successful at elevating the $NO_2^- + NO_3^-$ nutrient concentrations for enriched cinderblocks approximately fourfold relative to controls (4.3 ± 1.4 μM versus 0.8 ± 0.3 μM, respectively) (Mann–Whitney; $p = .05$). There was a similar trend for enriching PO_4 concentrations for the enriched cinderblocks (0.6 ± 0.2 μM) relative to controls (0.2 ± 0.08 μM) (Mann–Whitney; $p = .1$). There was ~80% more sediment observed on coral tissues for corals in the sediment addition treatment than on non-sediment addition corals (one-way ANOVA; $F_{7,72} = 164.22$, $p < .0001$). The sizes of wounds from snail predation averaged 32.8 ± 2.2 mm^2 and did not vary across the different predation treatment combinations (one-way ANOVA; $F_{3,39} = 1.22$, $p = .32$).

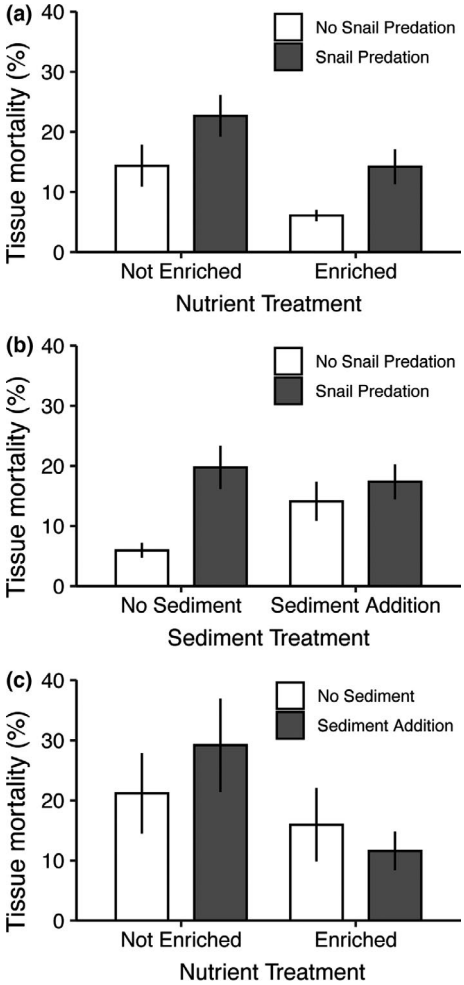

FIGURE 4 The percent tissue mortality (mean ± *SE*; $n = 10–11$ per treatment) of massive *Porites* nubbins after the addition of *Coralliophila violacea* snails. These data were averaged over three study dates (8/23, 8/30, and 9/6; refer to Figure S2). Panels are organized by the following significant two-way interactions (see Table 1 for results): (a) nutrient enrichment and snail predation, (b) sediment addition and snail predation, and (c) sediment addition and nutrient enrichment. Note the variable *x*-axes labels and legends on each panel

Tissue mortality on massive *Porites* fragments was relatively low before snail addition (0.3% overall) and increased after *Coralliophila violacea* were added (Figure S2). Five coral replicates, each from a different treatment combination, experienced complete mortality by the end of the experiment. There were no main effects of snail predation, nutrient enrichment, or sedimentation on tissue mortality

of corals (Table 1; Figure S3), although there was a trend for snail predation to increase tissue mortality (LMM; $F = 2.73$, $p = .1$). There was a significant two-way interaction between nutrient enrichment and snail predation where nutrients appeared to lower tissue mortality in corals, with the lowest mortality occurring when snail predation was absent (LMM; $F = 7.34$, $p < .01$; Figure 4a). We also observed a significant two-way interaction between snail predation and sedimentation (LMM; $F = 4.53$, $p = .034$; Figure 4b). In the absence of sediment, snail predation increased tissue mortality over threefold (10% without snails versus 31% with snails) while in the presence of sediment, tissue mortality was similar with or without snails (28% without snails versus 31% with snails). Lastly, there was a trend for nutrient enrichment to reduce the tissue mortality of sediment exposed corals (LMM; $F = 3.80$, $p = .053$; Figure 4c). Despite the significant two-way interactions, within-group multiple comparisons for these two-way interactions did not reveal any significant pairwise differences (all comparisons $p > .2$), likely due to low sample size. Despite the interactions of the main factors on coral tissue mortality, we did not see any impact of nutrients or sedimentation on the ability of corals to heal predation wounds from snails (Table 2; Figure 5).

Densities of Symbiodiniaceae were 28% higher in corals enriched with nutrients (2.68 ± 0.24 cells $\times 10^5$ cm^{-2}) than in corals without enrichment (2.09 ± 0.24 cells $\times 10^5$ cm^{-2}) (LMM; $F = 5.64$, $p = .021$; Table 3; Figure 6a). There were no other main or interactive effects of the treatments on Symbiodiniaceae densities (Table 3). We observed a trend in which chlorophyll *a* concentration per symbiont cell varied across nutrient enrichment treatments (LMM; $F = 3.95$, $p = .052$; Table 4; Figure 6b). We did not observe effects of snail predation or sediment addition on chlorophyll *a* concentration (Table 4).

4 | DISCUSSION

Our study investigated relationships among sedimentation, nutrient enrichment, and corallivory by the snail *Coralliophila violacea* across reefs and the consequences of these stressors to individual colonies of massive *Porites*. At the reef scale, nitrogen (N) enrichment and sediment accumulation on massive *Porites* colonies were negatively related across reefs, with sites higher in N availability having lower sediment accumulation on corals. Yet sites with higher N availability had higher densities of *C. violacea* on massive *Porites* colonies. Despite these contrasting patterns, *Porites* corals at sites with higher N, more snails, and less sedimentation had similar amounts of live tissue as sites with lower N, fewer snails, and more sedimentation. At the scale of coral colony, results from our experiment suggest that snail predation increased tissue mortality of massive *Porites*, but that sedimentation or nutrient enrichment did not exacerbate this mortality. In fact, sediment addition increased tissue mortality such that the impact of corallivorous snails was no longer evident, with no differences in tissue mortality between corals that were preyed upon and control corals under sediment addition. Nutrient enrichment appeared to interact with corallivory and sediment addition to reduce tissue mortality, possibly by facilitating increases in algal symbiont densities in corals. Thus, instead of these stressors acting synergistically to drive the decline of corals, predation, nutrients, and sedimentation appear to be antagonistic with one another, somewhat mitigating the presence of multiple stressors on coral physiology and mortality.

TABLE 2 Linear mixed-effects model results for the percent of each wound healed with Kenward–Roger approximation for degrees of freedom

Fixed effects	*df*	*F*	*p*
Nutrient	1	0.0326	.861
Sediment	1	0.0482	.829
Snail shell length	1	0.0200	.889
Nutrient × Wounded	1	0.884	.360

Notes: *p*-Values defined as significant at a threshold of .05 are highlighted in bold.

4.1 | Interactions of sedimentation and nutrients on coral mortality

Sediment accumulation and N loading covary negatively at the reef scale across our sites in Mo'orea. This pattern is surprising given that nutrient loading and sedimentation are often positively related as a product of land-based pollution (Lapointe et al., 2004; Lirman & Fong, 2007). In contrast to our findings, Becker and Silbiger (2020) showed that sedimentation rates and nutrient enrichment are positively correlated across fringing reefs that overlap our study sites during the Austral winter. The discrepancies between these findings may be explained by seasonal differences in sedimentation and nutrient regimes across these sites or by methodological differences; we measured the prevalence of sediment accumulation on coral colonies, not sedimentation rates. Yet the effects of these stressors on live coral cover remain equivocal. For example, simultaneous sedimentation and nutrient loading have been correlated with low coral cover in Indonesia (Edinger et al., 2000) but not in the Florida Keys Reef Tract (Lirman & Fong, 2007). Moreover, experimental evidence that these two stressors interact to have detrimental effects on corals is lacking and long-term nutrient enrichment may have more deleterious impacts to coral cover than sedimentation alone (Muthukrishnan & Fong, 2014).

The results from our field experiment, along with other studies (Gil et al., 2016; Muthukrishnan & Fong, 2014), suggest that the co-occurrence of sedimentation and nutrient enrichment rarely exacerbates coral tissue mortality. In fact, the co-occurrence of these stressors reduced tissue mortality in our experiment, supporting our hypothesis that nutrient enrichment would mitigate the harmful effects of sedimentation. It is worth noting that our experiment occurred on a shorter time scale (~6 weeks) than other studies documenting chronic negative effects of nutrient enrichment on live coral cover (Lapointe et al., 2019; Muthukrishnan & Fong, 2014). However, over the short term, nutrient enrichment appeared to benefit the massive *Porites* in our experiment. Edinger et al., (2000)

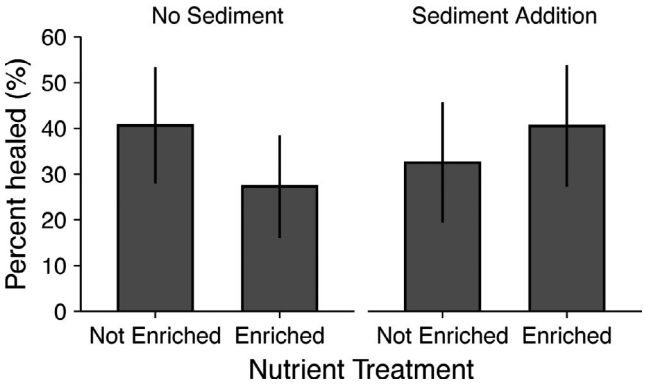

FIGURE 5 Percent of each wound healed (mean ± *SE*; *n* = 10–11 per treatment) for massive *Porites* nubbins exposed to predation by *Coralliophila violacea*. Corals were exposed to nutrient enrichment and sediment addition

TABLE 3 Linear mixed-effects model results for Symbiodiniaceae densities with Kenward–Roger approximation for degrees of freedom

Fixed effects	*df*	*F*	*p*
Predation	1	1.784	.186
Nutrient	1	5.643	**.0208**
Sediment	1	0.0015	.969
Snail shell length	1`	1.125	.293
Predation × Nutrient	1	0.0113	.916
Predation × Sediment	1	0.0899	.765
Nutrient × Sediment	1	0.249	.620
Predation × Nutrient × Sediment	1	0.636	.428

Notes: p-Values defined as significant at a threshold of .05 are highlighted in bold.

suggested that nutrient supply may enhance coral metabolism and enable corals to counteract the adverse effects of sedimentation, resulting in no observable impacts to coral condition (e.g., no changes to net growth). Our study provides some evidence in support of this hypothesis because increased nutrient supply mitigated partial mortality for massive *Porites* individuals under sedimentation stress, possibly by facilitating increased densities of algal symbionts. Given that nutrients have complex effects on corals depending on nutrient identity, coral identity, and nutrient concentration (Gil, 2013; Shantz & Burkepile, 2014), the moderate enrichment levels in our experiment may simply not be deleterious to this coral.

Even though we did not observe a synergistic effect between nutrients and sedimentation, sediment accumulation on corals can induce tissue paling and even tissue necrosis in some circumstances (Fabricius, 2005). In our field surveys, the accumulation of sediment on massive *Porites* was associated with paling tissue in ~80% of occurrences, suggesting that this is an important source of physiological stress for colonies in Mo'orea. In our experiment, sedimentation appeared to interact with corallivory where sedimentation alone caused as much tissue mortality as corallivory but did not exacerbate the impact of corallivory. Field evidence supports this, as massive

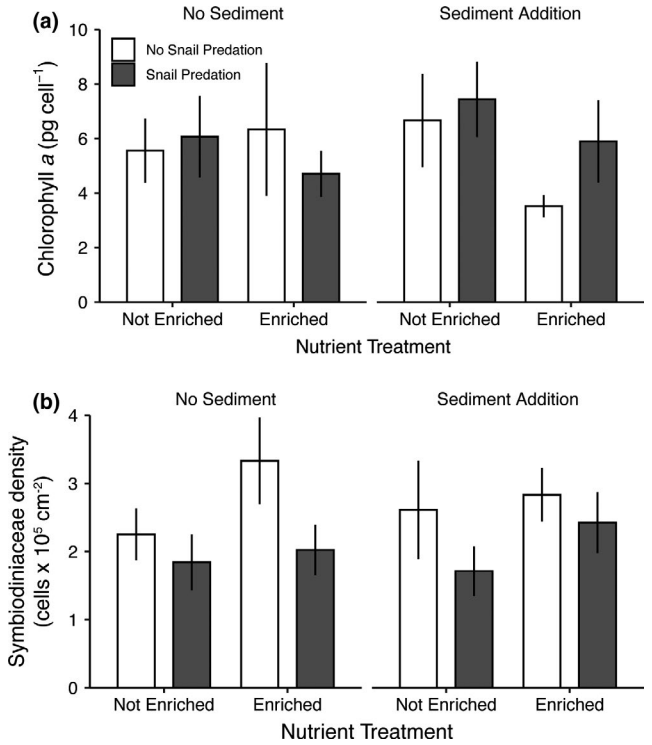

FIGURE 6 The (a) Symbiodiniaceae density (cells × 10^5 cm^{-2}; mean ± *SE*; *n* = 9–11 per treatment) and (b) Chlorophyll *a* concentration (pg $cell^{-1}$; mean ± *SE*; *n* = 9–11 per treatment) of massive *Porites* nubbins subjected to nutrient enrichment, sediment addition, and predation by *Coralliophila violacea* (white bars: no predation; gray bars: snail predation)

TABLE 4 Linear mixed-effects model results for chlorophyll *a* (pg $cell^{-1}$) with Kenward–Roger approximation for degrees of freedom

Fixed effects	*df*	*F*	*p*
Predation	1	0.299	.587
Nutrient	1	3.945	**.0520**
Sediment	1	0.265	.609
Snail shell length	1	0.220	.641
Predation × Nutrient	1	0.0149	.903
Predation × Sediment	1	0.776	.382
Nutrient × Sediment	1	0.345	.559
Predation × Nutrient × Sediment	1	0.244	.623

Notes: p-Values defined as significant at a threshold of .05 are highlighted in bold.

Porites cover in the nearshore reefs of Mo'orea has been negatively correlated with sedimentation rates (Gowan et al., 2014).

4.2 | Interaction between nutrients and corallivory on coral mortality

The higher densities of *Coralliophila violacea* observed at sites with more N loading may also play an important role in impacting massive

Porites colonies at the reef scale. Land-based pollution increases phytoplankton populations that often promote invertebrate larval success (Brodie et al., 2017; Fabricius et al., 2010; Sutcliffe Jr, 1973), which might, in turn, promote higher muricid gastropod population densities on coral reefs with higher nutrient loading (Li et al., 2013; Moyer et al., 1982). Populations of corallivorous snails can increase after sites are perturbed by land-based pollution, resulting in higher snail densities on live coral colonies (Li et al., 2013; Moyer et al., 1982). For instance, in Igaya Bay, Japan, the construction of a cattle pasture near river outlets coincided with a population explosion of *Drupella* spp. snails that decimated live coral cover (Moyer et al., 1982). Similarly, we observed higher snail densities at more enriched sites. However, it is worth noting that the snail densities in our field surveys are orders of magnitude lower (mean: 0.4 ind. M^{-2} live coral tissue at high N sites) than those observed in other studies in the Pacific (Moyer et al., 1982:3,000 ind. m^{-2}, Li et al., 2013:8.6 ind. m^{-2}). These lower densities may be one reason why we saw no differences in live coral tissue on reefs with higher nutrient loading and higher snail densities.

In our field experiment, snail predation increased tissue mortality, particularly when nutrient enrichment and sedimentation were not present. Similarly, predation by the snail *C. abbreviata* led to ~30% higher mortality for *Acropora cervicornis* (Lamarck, 1816) nubbins relative to corals without snails (Shaver et al., 2017). However, our experiment showed that nutrient enrichment reduced overall tissue mortality for massive *Porites* corals in both the absence and presence of snails. This pattern may explain why we did not see less live coral tissue at the reef scale where higher nutrient availability and increased corallivorous snail densities co-occurred. Even though nutrient enrichment reduced overall levels of tissue mortality, enrichment increased the magnitude of the effect of snails on tissue mortality as compared to the effect on corals that were not enriched (compare enriched versus not enriched corals in Figure 4a). Thus, nutrient supply made the impact of snail predation on tissue mortality more apparent relative to corals under ambient nutrient conditions.

The mitigating effect of nutrients on corallivory was somewhat surprising given that another recent study showed that corallivory interacted with excess nutrients to increase *Porites* mortality fivefold compared to corals without nutrients (Zaneveld et al., 2016). The nature of the corallivory may be an important distinction. Previous work showing negative interactions between nutrients and corallivory on corals have been with excavating corallivores, such as parrotfish, that remove coral tissue and coral skeleton (Zaneveld et al., 2016). However, our work and that of others (Shaver et al., 2017) showing no exacerbation of corallivory damage by nutrients have worked with browsing corallivorous snails that make less injurious wounds to corals by only removing coral tissue. These differences in the mode of corallivory may have a significant effect on how corals respond to corallivory and how nutrients impact these responses (reviewed by Rice et al., 2019).

Excess nutrients may also increase the ability of corals to withstand the negative effects of corallivory. Corals can respond positively (e.g., increased symbiont populations, chlorophyll production, and coral growth) to moderate increases in inorganic nutrients (Becker & Silbiger, 2020; Ezzat et al., 2015; Gil, 2013; Marubini & Davies, 1996). This effect might allow corals to compensate for the detrimental effects of predation. In our experiment, elevated nutrient supply increased the density of algal symbionts by 28%, potentially increasing the availability of carbon products for the host. Increased carbon products may enable corals to cope with concurrent stressors, especially corallivorous snails that typically impact corals by depleting carbon stores (Oren et al., 1998). This mechanism may be one reason why, at the reef scale, we saw no differences in the percent of live coral tissue of massive *Porites* colonies at low or high nitrogen sites. Thus, nutrient enrichment at the reef scale might be counteracting the effects snail corallivory by boosting algal symbiont physiology.

4.3 | Massive *Porites*, tissue healing, and abiotic factors

Based on previous findings, we reasoned that nutrient enrichment would accelerate wound healing (Rice, Maher, et al., 2019) and expected that sedimentation would impair wound healing (Cróquer et al., 2002). Contrary to these hypotheses, nutrient enrichment and sedimentation had no independent or interactive effects on the tissue regeneration rates of massive *Porites* in our field experiment. Other studies have documented that nutrient enrichment much higher than that in our treatments (Koop et al., 2001: ~11 μM ammonium, ~2 μM phosphate; Renegar et al., 2008: ~11 μM nitrate, ~4 μM phosphate; present study: ~4 μM nitrate, ~0.6 μM phosphate) and sediment (Cróquer et al., 2002: ~266 mg cm^{-2} day^{-1}; present study: ~100 mg cm^{-2} day^{-1}) hinder the tissue regeneration capacity of corals independently. The disparities in these patterns may be explained by several non-mutually exclusive mechanisms.

First, the thicker tissues characteristic of massive *Porites* (range: 4–8 mm thickness) (Linsley et al., 1999; Suzuki et al., 2005) and its perforate skeleton enable this coral to store more lipid reserves and to translocate resources across the colony surface more efficiently, enabling rapid wound healing. Second, *Porites* spp. corals in the present study and others (Marubini & Davies, 1996) had increased densities of Symbiodiniaceae under nutrient enrichment, which may increase the photosynthate products available to fuel tissue regeneration. Third, nutrients at concentrations similar to the enrichment treatments of the present study are assimilated by *Porites* tissues (Sims et al., 2020) and correlated with elevated protein in host and symbiont tissues (Marubini & Davies, 1996). Assimilated nitrogen from nutrient enrichment may increase the mineral nitrogen available for protein, subsequently increasing the protein available for new coral tissues to heal injuries. Fourth, we observed the complete clearing of added sediment within ~24 hr through wave action or mucus sheet production, which is a strategy used by massive *Porites* to shed sediment particles (Bessell-Browne et al., 2017). Our sediment treatment may not have had the required residency time to influence wound healing. Lastly, *Porites* has been shown to maintain

rapid tissue regeneration under other stressors (e.g., warmer temperatures) (Edmunds & Lenihan, 2010). Our results provide further evidence that this coral taxon is robust at maintaining tissue regeneration under various abiotic stressors. Taken altogether, our results indicate that the healing capacity of massive *Porites* may facilitate its success under a variety of abiotic templates and its tolerance to various anthropogenic stressors.

5 | CONCLUSIONS

Understanding the additive and interactive effects of multiple abiotic and biotic forces is crucial to predict how physical and ecological processes may interact to alter the physiology and mortality of species (Sala et al., 2000). Sedimentation and nutrient loading are common drivers on coral reefs that can increase coral mortality (Fabricius, 2005; Lapointe et al., 2019) and mediate how corals respond to biotic forces such as predation (reviewed by Rice, Maher, et al., 2019). Our study indicates that these multiple stressors interact in complex ways to drive changes to massive *Porites* mortality at the colony scale, which may explain the patterns in live coral tissue observed at the reef scale. For example, we did not observe differences in live coral tissue across survey sites despite the positive correlation between nitrogen loading and the density of corallivorous snails. This might be due to nutrient enrichment increasing algal symbiont densities at the colony level and reducing overall tissue mortality. Moreover, the effect of snail predation on coral mortality was suppressed by sedimentation. Nutrient supply also reduced overall coral mortality for corals exposed to either sedimentation or snail predation. Thus, these multiple stressors acted antagonistically, rather than synergistically, to drive changes to coral mortality and physiology in our factorial field experiment. This antagonism among multiple stressors appears to be common across ecosystems and types of stressors (Crain et al., 2008; Darling & Cote, 2008; Jackson et al., 2016; Maher et al., 2019). Given that many coral reefs experience multiple stressors simultaneously, future research focusing on the nature of synergisms or antagonisms of biotic and abiotic forces will be key to understanding the trajectory of coral reef communities in the face of increasing human-induced stressors.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

All code and raw data used for this study are available at https://github.com/mallorymrice/SNAIL_2017

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SUPPORTING INFORMATION

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